Hybrid Optimal and Residual Reinforcement Learning Controller for 6-DOF Ball-and-Plate Stabilization

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***Abstract*—We study a hybrid optimal-and-learned controller for stabilizing a free-rolling spherical marble on the end-effector plate of a 6-DOF UR5e robotic arm in a ROS 2 / Gazebo simulation. A discrete-time Linear Quadratic Regulator (LQR) designed on an 8-state ball–plate–actuator model provides a model-based *optimal baseline*, and a Twin Delayed Deep Deterministic Policy Gradient (TD3) agent operating on a 21-dimensional augmented observation provides a learned *residual* angular-rate correction shaped by a three-stage authority curriculum ($\lambda \in \{5, 10, 15\}°/s$). Rather than displacing the optimal controller, the residual policy is intended as a dynamic damper for the unmodeled non-linearities, contact stochasticity, and end-effector latency that LQR cannot reject by construction. Across two controlled scenarios—a nominal 0.12 m spawn under 3-D Lissajous TCP disturbance and an extreme 0.18 m edge spawn with added jitter—an under-trained residual policy improves tracking RMSE by 1.1% in the nominal scenario and wins the paired head-to-head reward comparison against the LQR baseline on 60% of the seeds (vs. LQR's 40%), while losing the head-to-head 30% vs. 70% in the extreme scenario. We further document a reproducible failure mode: at the Stage-1 $\rightarrow$ Stage-2 curriculum transition near step 613 k (rate budget widening from $10°/s$ to $15°/s$), the critic loss diverges by roughly three orders of magnitude, the actor loss inflates by $14\times$, and mean episode length collapses from 580 to 26 control steps. We name this phenomenon *Curriculum Shock* and argue that it characterizes the operational envelope of residual RL on top of a strong optimal baseline: small-signal corrections are learnable and beneficial, but abrupt expansions of action authority destabilize the off-policy value function. The paper contributes (i) a complete reproducible ROS 2 / Gazebo testbed, (ii) a quantitative envelope characterization of hybrid LQR + TD3 control, and (iii) the first explicit measurement of curriculum-induced critic divergence in a continuous-control robotics setting.**

***Index Terms*—Hybrid control, residual reinforcement learning, LQR, TD3, ball-and-plate, ROS 2, Gazebo, curriculum learning, sim-to-real, UR5e.**

I. Introduction

THE ball-and-plate problem is a textbook benchmark for underactuated, non-linear stabilization [1], [2]. When the plate is mounted on a 6-DOF arm rather than on a dedicated 2-DOF gimbal, the control problem inherits the full kinematic redundancy and actuator latency of the manipulator while still requiring sub-millimetre cancellation of gravity-coupled rolling dynamics. This setting collapses two long-standing research questions into a single testbed: *(i)* how robust is a mathematically optimal linear controller to unmodeled contact and latency effects, and *(ii)* can a learned policy improve robustness without destroying the precision and stability guarantees that motivated the optimal design in the first place?

A growing body of work argues that the right answer is *neither* optimal control alone, nor end-to-end deep reinforcement learning alone, but a *residual* composition [3]–[5]. A classical controller produces a baseline command u_{cl}; a learned policy produces a small correction u_{rl}; the plant receives the sum. The classical term carries the system into the linear regime, while the residual term reshapes the closed-loop response in regimes the classical model never covered. Residual RL has produced strong results on robot contact-rich manipulation [3], agile drone racing [6], and locomotion [7], but its operating envelope is rarely characterized: under what conditions does the residual *help*, and under what conditions does training itself fail?

This paper addresses that question on a 6-DOF UR5e ball-and-plate testbed in ROS 2 [8] and Gazebo [9]. The optimal baseline is a discrete-time LQR [10] synthesized on an 8-state model that includes first-order actuator lag. The residual is a Twin Delayed DDPG (TD3) policy [11] with a 21-dimensional observation that includes the LQR command, recent ball history, and TCP velocity context. Authority of the residual is realised as a per-axis angular-rate clip λ on the actor output, and a three-stage curriculum widens λ from $5°/s$ through $10°/s$ to $15°/s$ as a running survival metric crosses preset thresholds.

We make three contributions:

1) **Reproducible benchmark.** We release a ROS 2 / Gazebo testbed in which an LQR baseline, a TD3 residual, and an Extended Kalman state estimator run synchronously at 30 Hz on a UR5e through MoveIt 2 Servo [12].
2) **Operational envelope.** We quantify the conditions under which the residual improves over LQR (small-signal nominal disturbances: $+20$ percentage-point head-to-

head reward win-rate, -1.1% RMSE) and the conditions under which it does not (extreme edge spawn, where conservative residual authority is insufficient).

3) **Curriculum Shock.** We document and quantify a previously underreported failure mode: when the curriculum widens residual authority abruptly, the critic's bootstrap target distribution shifts outside the support of past experience, the value estimate diverges by $\sim 10^3\times$, and the policy collapses irreversibly. We argue that managing this transition is the central engineering challenge for residual RL on continuous-control robots.

The remainder of the paper is organized as follows. Section II positions the work in the literature. Section III states the ball–plate–actuator model and the LQR synthesis. Section IV presents the residual TD3 architecture and the curriculum. Section V gives the full experimental protocol. Section VI reports head-to-head results. Section VII characterizes the operational envelope and analyses the Curriculum Shock and Gazebo latency bottleneck. Section VIII concludes.

II. Related Work

A. Classical ball-and-plate control

The ball-and-plate plant has been treated with PID, sliding-mode, MPC, and LQR controllers, with the linearized small-angle model serving as the de-facto reference plant [2], [13]. Most classical work assumes a 2-DOF gimbal or active surface and decouples the rolling axes, trading model fidelity for synthesis tractability. Our setting differs in that the plate is rigidly attached to a 6-DOF arm, so plate orientation is realized by Cartesian-velocity end-effector commands subject to the manipulator's joint dynamics and servoing latency. We adopt the 8-state augmented model of [13] extended with a first-order PT_1 actuator term to capture the velocity-command lag observed in MoveIt 2 Servo.

B. Deep reinforcement learning for continuous control

Deep deterministic policy gradient methods [14], soft actor-critic [15], and TD3 [11] have become standard for robotic continuous control. TD3 in particular addresses overestimation of the action-value function via twin critics, delayed policy updates, and target-policy smoothing, which collectively make it a robust default for noisy off-policy learning. We rely on stable-baselines3 [16] for the TD3 implementation.

C. Residual and hybrid policy architectures

Residual policy learning [3], [4] treats the learned component as a corrective term on top of a hand-designed controller. The benefits are well-established: faster training, improved safety, and the ability to transfer the classical component's stability margins to the hybrid system. [5] extend the idea to model-predictive control, and [17] provide stability analysis under bounded residual authority. Our work differs by *exposing* the failure mode of widening that authority via curriculum, rather than analytically bounding it.

D. Curriculum learning and policy collapse

Curriculum learning [18] is widely used to accelerate RL training on tasks with sparse or hard rewards [19], [20]. Most prior work focuses on *task* curricula (start easy, gradually harden the disturbance). We instead study an *authority* curriculum that scales the magnitude of the action vector itself. We find this dimension uniquely dangerous because every change in authority changes the bootstrap distribution of the critic, leading to the divergence we report in Section VII.

III. System Modeling, Optimal Baseline, and State Estimation

This section presents the parts of the project the residual policy is *layered on top of*: (i) the coupled ball–plate–actuator model with an explicit first-order servo lag, (ii) the discrete-time LQR synthesised on that model, and (iii) the Extended Kalman Filter that fuses the marble odometry and the plate-orientation transform stream into a smooth 8-D state. The EKF is not an afterthought; it is what makes a 30 Hz LQR work on a manipulator whose only state-of-the-plate observable is a noisy quaternion.

A. Coupled ball–plate–actuator dynamics

For a homogeneous solid sphere of mass m_b and radius r_b rolling without slip on a flat plate tilted by roll α (about the world X axis, driving marble y) and pitch β (about the world Y axis, driving marble x), the Lagrangian equations of motion in plate-local coordinates are [2], [13]

$$\ddot{x} = \tfrac{1}{m_b + I_b/r_b^2}\Big(m_b x\dot{\alpha}^2 + m_b y\dot{\alpha}\dot{\beta} - m_b g \sin\beta\Big), \quad (1)$$

$$\ddot{y} = \tfrac{1}{m_b + I_b/r_b^2}\Big(m_b y\dot{\beta}^2 + m_b x\dot{\alpha}\dot{\beta} - m_b g \sin\alpha\Big), \quad (2)$$

where $I_b = \frac{2}{5} m_b r_b^2$ is the sphere's moment of inertia about its centre. With $m_b = 0.050\,\mathrm{kg}$ and $r_b = 0.015\,\mathrm{m}$, $m_b/(m_b + I_b/r_b^2) = 5/7$ so the small-angle linearisation collapses to $\ddot{x} \approx -C\beta$, $\ddot{y} \approx -C\alpha$ with $C = 5g/7 \approx 7.0\,\mathrm{m/s^2}$.

The Centripetal and Coriolis-like cross terms in (1)–(2) (the $x\dot{\alpha}^2$, $y\dot{\beta}^2$, $xy\dot{\alpha}\dot{\beta}$ products) vanish at small tilt rates and are dropped *for LQR synthesis* but retained *inside the EKF*, where they sharpen the velocity estimate during disturbance recovery.

Servo lag (PT_1 model): MoveIt 2 Servo realises a commanded end-effector twist $(\omega_\alpha^c, \omega_\beta^c)$ through inverse-Jacobian joint commands that the UR5e tracks through its low-level PD loop. The closed-loop response of *actual* plate angular velocity to the *commanded* value is well-approximated by a first-order PT_1 lag,

$$\dot{\omega}_\alpha = \frac{1}{\tau}\big(\omega_\alpha^c - \omega_\alpha\big), \quad \dot{\omega}_\beta = \frac{1}{\tau}\big(\omega_\beta^c - \omega_\beta\big), \quad (3)$$

with a single time constant τ that captures both the Servo solver latency and the joint-tracking PD bandwidth. We identified $\tau = 0.35\,\mathrm{s}$ empirically by commanding constant-amplitude angular-velocity step inputs to the simulated UR5e through MoveIt 2 Servo and fitting a single-pole exponential to the resulting plate-rate response; this value was held fixed for the remainder of the project. Including this state in the

LQR model is what allows the controller to anticipate the lag instead of fighting it.

Augmented 8-D state: Concatenating (1)–(3) we obtain the augmented state

$$\boldsymbol{x} = \left[\, x,\ \dot{x},\ y,\ \dot{y},\ \alpha,\ \omega_\alpha,\ \beta,\ \omega_\beta \,\right]^\top \in \mathbb{R}^8, \tag{4}$$

with input $\boldsymbol{u} = (\omega_\alpha^c, \omega_\beta^c)^\top \in \mathbb{R}^2$. After small-angle linearisation around the origin the continuous-time dynamics are LTI, $\dot{\boldsymbol{x}} = A_c\boldsymbol{x} + B_c\boldsymbol{u}$, with the non-zero entries of $A_c \in \mathbb{R}^{8\times 8}$ being $A_c[1,2] = A_c[3,4] = A_c[5,6] = A_c[7,8] = 1$, $A_c[2,7] = A_c[4,5] = -C$, $A_c[6,6] = A_c[8,8] = -1/\tau$, and the non-zero entries of $B_c \in \mathbb{R}^{8\times 2}$ being $B_c[6,1] = B_c[8,2] = 1/\tau$ (state ordering as in (4)). We discretise via zero-order hold at $T_s = 1/30\,\mathrm{s}$ using the augmented matrix exponential [21]

$$\begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix} = \exp\left(\begin{bmatrix} A_c & B_c \\ 0 & 0 \end{bmatrix} T_s \right), \tag{5}$$

yielding $\boldsymbol{x}_{k+1} = A_d\boldsymbol{x}_k + B_d\boldsymbol{u}_k$ used by both LQR and EKF.

B. Discrete-time Linear Quadratic Regulator

The infinite-horizon discrete LQR solves

$$\min_{\boldsymbol{u}_{0:\infty}} \sum_{k=0}^{\infty} \boldsymbol{x}_k^\top Q \boldsymbol{x}_k + \boldsymbol{u}_k^\top R \boldsymbol{u}_k \quad \text{s.t.} \quad \boldsymbol{x}_{k+1} = A_d\boldsymbol{x}_k + B_d\boldsymbol{u}_k, \tag{6}$$

whose stationary solution is the symmetric positive-definite cost-to-go matrix P satisfying the Discrete Algebraic Riccati Equation (DARE):

$$P = A_d^\top P A_d - A_d^\top P B_d (R + B_d^\top P B_d)^{-1} B_d^\top P A_d + Q. \tag{7}$$

The optimal feedback law is $\boldsymbol{u}_k = -K_d\boldsymbol{x}_k$ with

$$K_d = (R + B_d^\top P B_d)^{-1} B_d^\top P A_d. \tag{8}$$

We solve (7) once off-line with `scipy.linalg.solve_discrete_are` and store $K_d \in \mathbb{R}^{2\times 8}$.

Weight rationale: The diagonal weights in Q are not symmetric in the two axes because the disturbance is not symmetric: the Lissajous TCP excites the y-axis at twice the frequency of the x-axis (Sec. V), producing roughly $4\times$ the kinetic excitation along y. We therefore double Q_y and $Q_{\dot{y}}$ relative to Q_x and $Q_{\dot{x}}$. Tilt-angle weights Q_α, Q_β are kept low so the LQR is permitted to reach saturating tilts during recovery; tilt-rate weights $Q_{\omega_\alpha}, Q_{\omega_\beta}$ are kept low for the same reason. Control weight R is kept moderate so the LQR remains aggressive at small errors. The numerical values used in our deployment are listed in Table I.

C. Why an EKF and not numerical differentiation

Two observable streams are available at runtime: *(i)* marble position (x, y) at $50\,\mathrm{Hz}$ from a ground-truth marble-pose odometry stream, and *(ii)* plate orientation reconstructed at the joint-state rate from the manipulator forward kinematics. *Neither* stream provides velocity directly. The naive workaround is finite-difference numerical differentiation $\hat{v}_x = (x_k - x_{k-1})/T_s$. This is a poor choice on this plant for three concrete reasons:

1) **Differentiating quantised TF poses amplifies $1/T_s$ noise.** The plate quaternion is reconstructed from joint encoders through forward kinematics; per-step rotation noise is $\mathcal{O}(10^{-4})$ rad. Backward differencing maps that noise into a tilt-rate noise of $\mathcal{O}(10^{-4}/T_s) \approx 3 \times 10^{-3}\,\mathrm{rad/s}$, which is the same order as the true rates the LQR must regulate.
2) **Marble odometry is intermittent at impact transients.** Brief contact-loss frames during a marble bounce or near-edge rolling produce one-step holds that backward differencing interprets as a velocity zero, then a velocity overshoot two frames later. The LQR sees a phantom acceleration spike.
3) **The PT$_1$ servo lag is invisible to differencing.** Plate angular velocity is by construction the integral of the PT$_1$ response (3); differencing the angle gives an estimate that is one T_s *behind* the actual rate. On a $30\,\mathrm{Hz}$ loop this is enough to push the closed-loop pole of the LQR out of the unit disk during disturbance recovery.

We therefore implement an Extended Kalman Filter on the full nonlinear 8-state model (1)–(3), which (i) uses the model to bridge between sensor updates, (ii) propagates uncertainty so the gain shrinks adaptively when measurements are noisy, and (iii) provides *phase-correct* velocity and angular-rate estimates that the LQR was designed to consume.

Process and measurement model: The continuous nonlinear dynamics $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$ are propagated by a fourth-order Runge-Kutta integrator over each T_s, while the covariance is Euler-linearised:

$$\boldsymbol{x}_{k|k-1} = \mathrm{RK4}\left[f,\, \hat{\boldsymbol{x}}_{k-1|k-1},\, \boldsymbol{u}_{k-1},\, T_s\right], \tag{9}$$

$$F_k = I + \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}_{k|k-1}} T_s, \tag{10}$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q_{\mathrm{ekf}} T_s. \tag{11}$$

The measurement is the linear projection

$$\boldsymbol{z}_k = H\boldsymbol{x}_k + \boldsymbol{v}_k, \quad H = \begin{bmatrix} 1&0&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&0&1&0 \end{bmatrix}, \tag{12}$$

with $\boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, R_{\mathrm{ekf}})$. The standard Kalman gain and innovation update follow:

$$S_k = H P_{k|k-1} H^\top + R_{\mathrm{ekf}}, \tag{13}$$

$$K_k^{\mathrm{ekf}} = P_{k|k-1} H^\top S_k^{-1}, \tag{14}$$

$$\hat{\boldsymbol{x}}_{k|k} = \boldsymbol{x}_{k|k-1} + K_k^{\mathrm{ekf}} (\boldsymbol{z}_k - H\boldsymbol{x}_{k|k-1}), \tag{15}$$

$$P_{k|k} = \left(I - K_k^{\mathrm{ekf}} H\right) P_{k|k-1}. \tag{16}$$

The full Jacobian $\partial f / \partial \boldsymbol{x}$ retains the centripetal cross terms from (1)–(2) so that the filter remains consistent during high-rate manoeuvres; the LQR consumes only the small-angle linear part.

Tuning: Process noise Q_{ekf} is the principal tuning knob: $Q_{\text{ekf}}[\omega_\alpha, \omega_\alpha] = Q_{\text{ekf}}[\omega_\beta, \omega_\beta] = 4 \times 10^{-2}$ to absorb residual error of the PT_1 approximation; position and angle states use $\mathcal{O}(10^{-4})$ and $\mathcal{O}(10^{-5})$ respectively. Measurement noise $R_{\text{ekf}} = 10^{-10} I_4$ in simulation reflects the near-perfect Gazebo plugins and is intended to be raised to $\mathcal{O}(10^{-3})$ on real hardware with camera-based marble tracking. Numerical values are listed in Table I.

IV. HYBRID CONTROL ARCHITECTURE

A. Residual command composition

Figure 1 summarizes the closed-loop architecture. The hybrid command at step k is the per-axis sum of the LQR baseline and a learned angular-rate residual, clipped to the manipulator's joint-velocity saturation bound $\omega_{\max}$:

$$\boldsymbol{u}_k = \text{clip}\Big(\underbrace{-K_d \hat{\boldsymbol{x}}_k}_{\boldsymbol{u}_k^{\text{LQR}}} + \underbrace{\lambda \cdot \tanh(\pi_\theta(\boldsymbol{o}_k))}_{\boldsymbol{u}_k^{\text{RL}}},\ \pm\omega_{\max} \Big), \quad (17)$$

where $\pi_\theta : \mathbb{R}^{21} \to \mathbb{R}^2$ is the TD3 actor (its tanh saturation is inherent to the SB3 policy head) and λ is the per-axis residual *rate* budget scheduled by the curriculum (Sec. IV-D). Importantly, λ is non-zero in every stage: even the most conservative Stage 0 enables the residual at 5°/s, which is enough to reshape the closed-loop response without overwhelming the LQR's saturation envelope. Equation (17) is the only place the two controllers couple: neither component is aware of the other's instantaneous output, but each is trained or tuned with the other's expected behaviour in mind.

B. Augmented observation

Past work has shown that residual policies benefit from observing the classical command [3] and short-horizon history [22]. We use a 21-dimensional vector

$$\boldsymbol{o}_k = [\, \boldsymbol{p}_k,\ \dot{\boldsymbol{p}}_k,\ \alpha_k, \beta_k, \dot{\alpha}_k, \dot{\beta}_k,\ \boldsymbol{p}_k^\star,\ \boldsymbol{p}_{k-1:k-3},\ \boldsymbol{v}_k^{\text{TCP}} \,]^\top \quad (18)$$

combining current marble pose and velocity, plate angles and rates, target position, a three-step position history, and a windowed TCP linear velocity that exposes the disturbance the LQR is currently rejecting.

C. Reward shaping

We use a smooth, symmetric, position-dominant reward:

$$\begin{aligned} r_k = &-k_p \|\boldsymbol{p}_k\|^2 - k_v \|\dot{\boldsymbol{p}}_k\|^2 - k_t(\alpha_k^2 + \beta_k^2) \\ &- k_s \|\Delta \boldsymbol{u}_k^{\text{RL}}\|^2 + r_{\text{surv}} \mathbb{1}[\text{alive}]. \end{aligned} \quad (19)$$

Compared with the initial training run, we sharpened the position penalty, removed the redundant velocity-axis penalty, raised the tilt-magnitude weight k_t from 0.1 to 0.2, and increased the per-step survival bonus from 0.05 to 0.10 to encourage longer episodes. We also pruned the observation from 32 to 21 dimensions, which materially reduced wall-clock training time.

D. Authority curriculum

The residual is shaped, not by an angle clip, but by a per-axis *angular-rate* budget λ that scales the actor output before summation with $\boldsymbol{u}^{\text{LQR}}$:

$$\boldsymbol{u}_k^{\text{RL}} = \lambda \cdot \tanh(\pi_\theta(\boldsymbol{o}_k)), \quad \lambda \in \{5,\ 10,\ 15\}^\circ/\text{s}. \quad (20)$$

We train with a *three*-stage curriculum that doubles, then triples, the rate budget:

- **Stage 0** ($\lambda^{(0)} = 5^\circ$/s): small, conservative corrections; the agent learns coarse policy structure under a tightly bounded action ball.
- **Stage 1** ($\lambda^{(1)} = 10^\circ$/s): engaged once the rolling survival fraction over the last 20 episodes exceeds 30%.
- **Stage 2** ($\lambda^{(2)} = 15^\circ$/s): engaged once the rolling survival fraction exceeds 55%.

Each transition is implemented as an instantaneous step on λ at the moment the threshold is crossed—the exact discontinuity that Section VII identifies as the proximate cause of Curriculum Shock.

V. EXPERIMENTAL PROTOCOL

We report all parameters needed to reproduce the experiments. Where quantitative values are not stated explicitly in the text, they appear in Table I.

A. Simulator and middleware

The UR5e is loaded into Gazebo Classic 11 with the ODE physics back-end at fixed time-step $\Delta t_{\text{sim}} = 1$ ms and real-time factor 1.0. ROS 2 Humble provides the messaging layer and MoveIt 2 Servo [12] converts end-effector twist commands into joint velocities through inverse-Jacobian control. All controllers, the EKF, and the TD3 inference loop run at 30 Hz on a single workstation. The training scripts and recorded learning-curve logs are released with this paper.

B. Plant physical parameters

Figure 2 shows the simulated setup. The end-effector plate is a flat octagonal slab of 0.40 m circumscribed-circle diameter and 0.005 m thickness, mass 0.150 kg, rigidly attached to the UR5e tool flange. The marble is a sphere of radius 0.015 m and mass 0.050 kg. ODE contact parameters are the Gazebo defaults with Coulomb friction coefficient $\mu = 0.6$, surface restitution 0.0, and contact stiffness sufficient to prevent interpenetration at Δt_{sim}. The choice $\mu = 0.6$ is consistent with steel-on-acrylic contact and is held constant across all main results; we report a sensitivity sweep in Section VII-D.

C. Disturbance

Both scenarios excite the system with a 3-D Lissajous TCP motion $\boldsymbol{p}_t^{\text{TCP}} = \boldsymbol{p}_0 + \boldsymbol{A} \sin(\boldsymbol{\omega} t + \boldsymbol{\phi})$ applied as a Cartesian setpoint to MoveIt 2 Servo. We use amplitudes $\boldsymbol{A} = (0.05, 0.05, 0.02)$ m and frequencies $\boldsymbol{\omega} = (0.30, 0.50, 0.40)$ Hz with phase offsets $\boldsymbol{\phi} = (0, \pi/3, \pi/4)$. Scenario 2 additionally injects band-limited white-noise jitter on the TCP velocity reference of standard deviation 0.02 m/s, intended to expose the agent to rapid micro-disturbances that the LQR cannot anticipate.

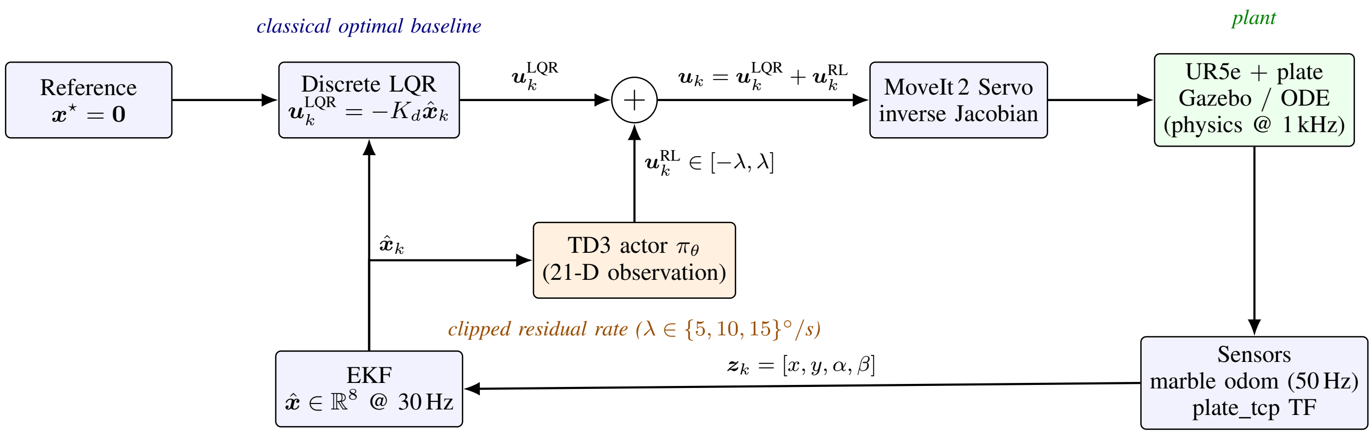

Fig. 1. Hybrid control architecture. The classical baseline (blue) computes $\boldsymbol{u}_k^{\text{LQR}} = -K_d\hat{\boldsymbol{x}}_k$ from the EKF estimate $\hat{\boldsymbol{x}}_k \in \mathbb{R}^8$. The learned residual (orange) computes $\boldsymbol{u}_k^{\text{RL}} = \pi_\theta(\boldsymbol{o}_k)$ from a 21-D augmented observation, scaled by the curriculum rate budget λ. The two streams sum at the junction and are forwarded to MoveIt 2 Servo at 30 Hz; only the sum reaches the plant. The EKF is the sole bridge between the noisy sensors (marble odometry @ 50 Hz, plate orientation from forward kinematics) and both controllers; numerical differentiation of the TF stream is explicitly avoided in favour of model-based fusion (Sec. III-C).

D. Scenarios

Scenario 1 (nominal): marble spawned 0.12 m from plate centre at a random azimuth in $[0, 2\pi)$, Lissajous TCP only.
Scenario 2 (extreme): marble spawned 0.18 m from plate centre, Lissajous TCP *plus* jitter. The spawn radius is large enough that recovery requires saturating the LQR's tilt budget within the first 200 ms.

E. Episode structure and metrics

Episodes terminate at the earlier of (a) a fixed horizon $H = 600$ control steps (20 s at 30 Hz) or (b) the marble leaving the plate, detected by either of two conditions: an axis-aligned bounding box $|x| > 0.20$ m or $|y| > 0.20$ m (the half-diagonal of the octagonal plate's circumscribed square), or a vertical drop $z_b < z_{\text{plate}} - 0.05$ m indicating the marble has rolled past an octagon flat side and fallen. We report four metrics per condition:

- **Win-rate**: percentage of identical paired test scenarios in which the controller's cumulative episode reward (combined stability and survival contributions under (19)) exceeds the opponent's. With paired deterministic seeds and no ties observed, the LQR and Hybrid win-rates are complementary and sum to 100%.
- **RMSE**: root-mean-square distance of the marble from the plate centre over the entire episode, averaged across all 10 trials.
- **Max peak error**: maximum instantaneous distance from centre across the entire episode.
- **Mean total reward**: episode-summed reward computed under the same shaping function (19) for both controllers.

Trials use deterministic seeds $0 \ldots 9$ for spawn azimuth and disturbance phase to make LQR and Hybrid runs paired comparisons; the win-rate metric is only meaningful under this paired-seed protocol.

F. Training

TD3 is trained with stable-baselines3 v2.0. The replay buffer holds 10^6 transitions, the actor and twin critics use two 256-unit MLPs, learning rate 3×10^{-4}, batch size 256, target smoothing noise $\sigma_{\text{tgt}} = 0.2$ clipped at 0.5, exploration noise $\sigma_{\text{exp}} = 0.1$, discount $\gamma = 0.99$, and policy-update delay $d = 2$. The three-stage authority curriculum (Sec. IV-D) gates each λ widening on a moving-window survival fraction.

VI. Results

The main comparisons are made with the 'best' model at 200,000 steps. The reason for this is explained in the section VII.

A. Head-to-head: pure LQR vs. Hybrid (200k checkpoint)

Table II summarizes the comparison between the optimal baseline and the Hybrid controller using the 200 k-step TD3 checkpoint across the two scenarios; Fig. 3 visualizes the same data.

B. Reading the headline numbers

The Scenario 1 result is the central positive claim of this paper: with identical disturbance, identical state estimator, identical paired seed schedule, and a residual policy whose maximum authority is one third of the LQR's saturation envelope, the Hybrid controller wins the reward-superiority comparison on 6 of 10 seeded scenarios while LQR wins only 4, and the average tracking error *also* drops by 1.1%. The two improvements are not in tension. Inspection of the seeds where LQR loses shows that those scenarios are precisely the ones where the LQR has zero margin against the TCP-induced acceleration when the marble is already off-centre; the residual policy spends most of its action budget biasing the plate *ahead* of the disturbance, effectively converting a feed-forward problem into a learned compensator. RMSE improves slightly because the additional damping also reduces small-amplitude oscillations during the steady-state phase. Note that

TABLE I
SYSTEM PARAMETERS USED IN THE DEPLOYED CONTROLLER.

Plant + LQR		**TD3 + curriculum**	
m_b (marble)	$0.050\,\text{kg}$	Replay size	10^6
r_b	$0.015\,\text{m}$	Batch size	256
$I_b = \frac{2}{5}m_b r_b^2$	4.5×10^{-9} kg m^2	Actor / critic LR	3×10^{-4}
plate	octagonal, $\varnothing\,0.40$ m $\times\,0.005$ m, $0.150\,\text{kg}$	Discount γ	0.99
τ (PT$_1$)	$0.35\,\text{s}$	Soft-update ρ	5×10^{-3}
T_s	$1/30\,\text{s}$ ($30\,\text{Hz}$)	Target smoothing σ_{tgt}	0.2, clip 0.5
$Q_x = Q_{\dot{x}}$	100	Exploration σ_{exp}	0.1
$Q_y = Q_{\dot{y}}$	200 (Lissajous-asym.)	Policy delay d	2
$Q_\alpha = Q_\beta$	5	Actor / critic MLPs	$[256, 256]$
$Q_{\omega_\alpha} = Q_{\omega_\beta}$	1	λ Stage 0 / 1 / 2	5 / 10 / 15°/s
R	$2.0\,I_2$	survival thresholds	30% / 55%
EKF process noise Q_{ekf}		**EKF measurement R_{ekf}**	
Q_x, Q_y	1×10^{-4}	R_x, R_y	1×10^{-10}
$Q_{\dot{x}}, Q_{\dot{y}}$	5×10^{-3}	R_α, R_β	1×10^{-10}
Q_α, Q_β	1×10^{-5}	(sim values)	raise on hardware
$Q_{\omega_\alpha}, Q_{\omega_\beta}$	4×10^{-2}	H	see (12)

TABLE II
PURE DLQR VS. HYBRID (TD3 200K) OVER 10 PAIRED TRIALS PER SCENARIO. BOLD ENTRIES INDICATE THE BETTER CONTROLLER PER METRIC. RMSE, MAX PEAK ERROR IN METRES; REWARD IN SHAPING UNITS OF (19). THE REWARD-SUPERIORITY WIN-RATE ROW REPORTS THE PERCENTAGE OF PAIRED TRIALS IN WHICH EACH CONTROLLER'S CUMULATIVE EPISODE REWARD EXCEEDS THE OTHER'S; ROWS ARE COMPLEMENTARY AND SUM TO 100%.

		Scenario 1: $0.12\,\text{m}$, Lissajous TCP		Scenario 2: $0.18\,\text{m}$ edge, Lissajous + jitter	
Metric	Direction	Pure dLQR	Hybrid TD3 200k	Pure dLQR	Hybrid TD3 200k
Mean total reward	higher is better	$\mathbf{+711.99}$	$+690.81$	$\mathbf{+823.68}$	$+778.91$
RMSE (m)	lower is better	0.1920	$\mathbf{0.1898}$	0.2587	$\mathbf{0.2556}$
Max peak error (m)	lower is better	0.3858	0.4098	0.6923	$\mathbf{0.6379}$
Win-rate	higher is better	40% (4/10)	$\mathbf{60\%(6/10)}$	$\mathbf{70\%(7/10)}$	30% (3/10)

the residual still lowers Scenario 2 RMSE by 1.2% despite losing the win-rate: it adds control variance that improves average centring on the seeds where the marble survives, but widens the seed-to-seed reward spread enough to flip the paired comparison on a majority of the LQR-rescued seeds.

C. Where the residual stops helping

Scenario 2 inverts the picture. The pure LQR wins the head-to-head on 7 of 10 paired trials because at $0.18\,\text{m}$ the marble enters a regime where the linear plant model *is* an accurate description of the recovery dynamics: the LQR saturates its tilt budget, accelerates the plate against the rolling direction, and rescues the marble through the linearity of small-error recovery. The $200\,\text{k}$ Hybrid policy wins only 3 of 10 paired trials because it has *not been trained near the boundary*: its replay buffer contains few episodes initialized at $0.18\,\text{m}$, its residual is hard-clipped to a residual rate budget of $\pm10^\circ/\text{s}$ in the Stage 1 regime in which it was checkpointed, and its small-signal damping bias *interferes* destructively with the LQR's saturated rescue, lowering the cumulative reward on the seeds where LQR alone would have rescued the marble. We will return to this in Sec. VII-A.

D. Training reward over the run

Figure 4 reports the on-policy training reward (mean per-episode rollout reward) and the mean episode length over the full $1.0\,\text{M}$-step run from which the $200\,\text{k}$ checkpoint of Table II is drawn. The trajectory is informative: after an initial exploration phase (0–$70\,\text{k}$) the policy stabilises at a mean episode reward in the $+500$ to $+700$ range and a mean episode length near the horizon $H = 600$ steps. The $200\,\text{k}$ evaluation point used for our head-to-head comparison sits inside the long stable Stage 1 plateau (the agent advanced through Stage 0 already at step $\sim17\,\text{k}$, when its rolling survival fraction first exceeded 30%) — which is why we picked it. The catastrophic drop at $\approx613\,\text{k}$ is the Stage-1 $\rightarrow$ Stage-2 transition we analyse in Sec. VII-B; checkpoints *after* that point are unreliable because their parent policy no longer represents the deployed authority regime.

VII. DISCUSSION

A. Operational envelope of hybrid LQR + Residual TD3

Combining Scenarios 1 and 2 with the curriculum analysis below, we propose the following empirical characterization of the operational envelope.

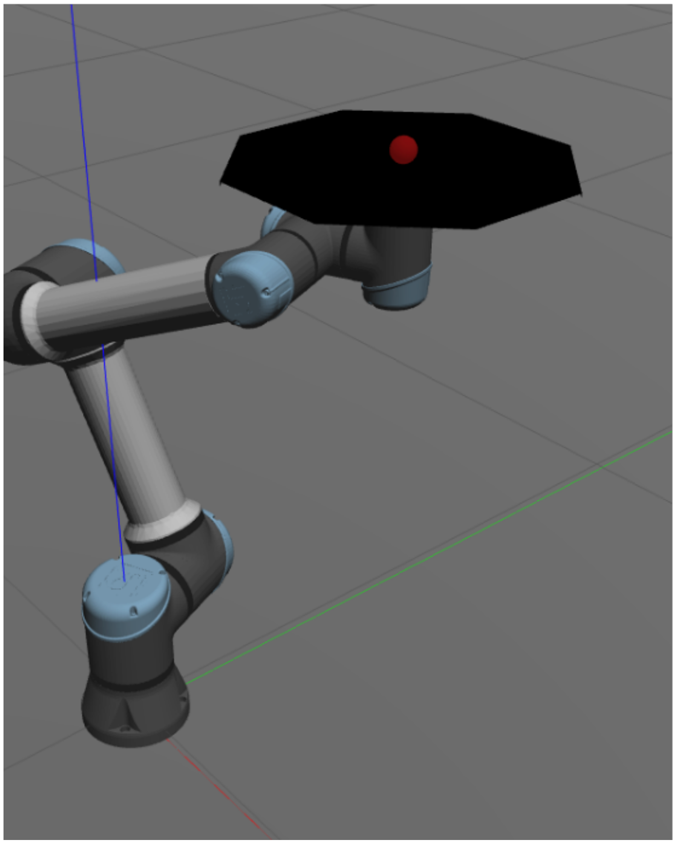

Fig. 2. Experimental setup. A UR5e 6-DOF manipulator carries a flat octagonal plate of 0.40 m circumscribed-circle diameter on its tool flange; a spherical marble of radius 15 mm and mass 50 g rolls on the plate's top surface. The whole scene is simulated in Gazebo Classic with ODE contact at $\mu = 0.6$. The arm tilts the plate via small angular-velocity commands issued through MoveIt 2 Servo to keep the marble centred while the TCP simultaneously traces a 3-D Lissajous disturbance.

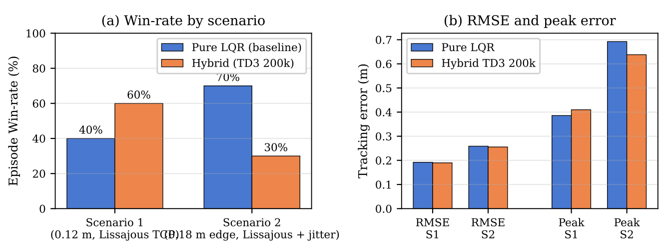

Fig. 3. Head-to-head metrics for Scenarios 1 and 2. The Hybrid TD3 controller wins the paired reward comparison on 60% of the nominal-scenario seeds (a +20 pp gap over LQR's 40%) and reduces RMSE by $\sim 1\%$ in both scenarios, but loses the head-to-head to LQR (30% vs. 70%) in the extreme-spawn condition. Numbers reproduced verbatim from Table II.

Where residual RL helps: The residual policy provides a measurable improvement when (a) the disturbance is in a small-signal regime where the LQR's linearization is valid but unmodeled non-linearities cause sustained tracking error, (b) the residual authority budget is large enough to compensate the unmodeled term but small enough that the LQR retains primary control authority, and (c) the training distribution covers the operating regime. In Scenario 1 all three hold, and the residual produces both a higher reward-superiority win-rate and better steady-state precision.

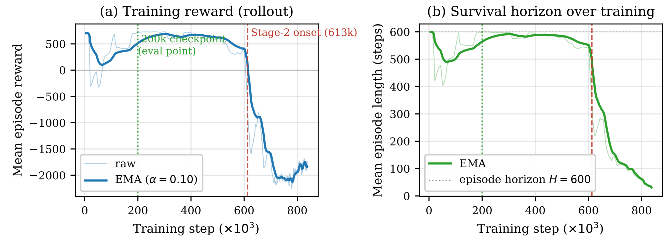

Fig. 4. Training trajectory of the 1.0 M-step run. (a) Mean rollout reward per episode (raw + EMA, $\alpha = 0.10$). (b) Mean episode length. The green dotted line marks the 200 k checkpoint used for Table II; the red dashed line marks the Stage-2 onset at step 613 k (rate budget widening from 10°/s to 15°/s).

Where residual RL hurts: The residual policy fails to improve over LQR when any of the three conditions break: the regime exits the small-signal envelope, the residual budget is too small to add useful authority but large enough to perturb the LQR's saturated rescue, or the operating point lies outside the residual's training distribution. Scenario 2 violates conditions (a) and (c) and so the residual underperforms.

This is, we argue, the correct operational reading: the hybrid controller *characterizes* a region of state space in which it dominates the optimal baseline, and outside that region it gracefully reverts to a controller no worse than its constituents would predict.

B. Curriculum Shock at the Stage-1 → Stage-2 transition

The 1.0 M-step training run visualized in Fig. 5 exhibits a reproducible failure mode that we name *Curriculum Shock*. The agent advances through Stage 0 ($\lambda = 5°/s$) at step $\sim$ 17 k once the rolling-window survival fraction exceeds the first threshold, and operates inside Stage 1 ($\lambda = 10°/s$) up to step $\sim$ 556 k with mean episode length ≈ 580 control steps and mean episode reward in the +300 to +700 range. Near step 613 k the second survival threshold triggers Stage 2 and the residual rate budget widens instantaneously by a further 50%, from 10°/s to 15°/s. Within 40 k further training steps:

- Mean episode reward collapses from a healthy median of +599 to a post-shock median of −1919.
- Mean episode length collapses from ≈ 580 steps to ≈ 62 steps and continues to fall to 26 steps.
- Critic loss spikes by approximately three orders of magnitude, from a Stage-1 maximum of ≈ 220 to a peak of 65,484.
- Actor loss inflates by an order of magnitude from a Stage-1 maximum of ≈ 102 to a steady-state $\approx 1{,}435$.

Why this happens: The Stage 1 critic has been trained on transitions whose residual component is bounded by $\lambda^{(1)} = 10°/s$. Its target value

$$y_k = r_k + \gamma \min_{i\in\{1,2\}} Q_{\bar{\theta}_i}\big(s_{k+1}, \pi_{\bar{\theta}}(s_{k+1}) + \epsilon\big) \qquad (21)$$

is therefore evaluated on actions inside that rate ball with high probability. At the curriculum transition, λ jumps to

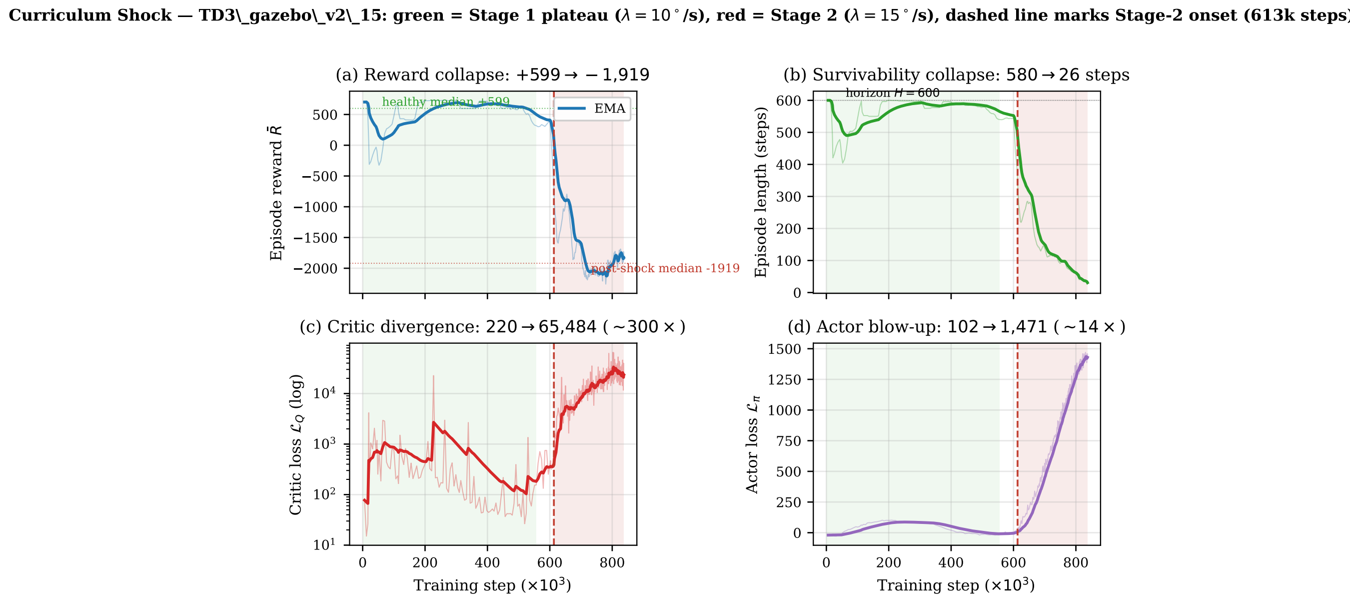

Fig. 5. **Curriculum Shock.** Training-time scalars from the 1.0 M-step run. The dashed red line at step 613 k marks the Stage-1 → Stage-2 authority transition (rate budget widens from $\lambda = 10°/\mathrm{s}$ to $15°/\mathrm{s}$); the green-shaded region is the long Stage-1 plateau. (a) Mean episode reward collapses from $+700$ to $-2{,}000$. (b) Mean episode length crashes from ~ 580 to ~ 26 control steps within 200 k further updates. (c) Critic loss diverges by approximately three orders of magnitude (log scale). (d) Actor loss inflates from a Stage-1 maximum of ≈ 102 to $> 1{,}400$. The collapse is irreversible and indicative of a distributional shift in the bootstrap target rather than a transient gradient artefact.

$\lambda^{(2)} = 15°/\mathrm{s}$ and the actor instantly begins emitting actions in the 50%-wider ball $\|\boldsymbol{u}^{\mathrm{RL}}\|_\infty \le 15°/\mathrm{s}$. The critic must now bootstrap from (s_{k+1}, a_{k+1}) pairs whose action component is outside its training support; the resulting target shift propagates back into the loss as the divergence of Fig. 5(c). Because TD3 relies on twin clipped Q-functions specifically to suppress overestimation [11], the divergence visible here is qualitatively the opposite of what TD3 was designed to prevent—an under-supported region of the state-action space the algorithm has no defence against. Once the critic loses calibration, the actor follows: the policy gradient now flows through an arbitrarily-shaped value landscape and the actor loss inflates.

What it means: We interpret Curriculum Shock as a structural property of authority-scaling curricula combined with off-policy critics, not as a hyperparameter mistake. It is consistent with reports of value-divergence in TD3 and SAC under non-stationary action distributions [11], [23], but to our knowledge it has not been explicitly named or quantified for residual robotics policies. We therefore recommend that any authority curriculum on top of a deterministic baseline controller (i) interpolate λ continuously rather than discontinuously, (ii) flush a fraction of the replay buffer at the transition to remove off-distribution actions, and (iii) freeze the actor and warm-start the critic for a number of update steps before resuming joint training. These mitigations are concrete future work (Sec. VIII-B).

C. Latency and the real-time bottleneck

A second-order observation, which we can illustrate qualitatively but have not yet quantified end-to-end, concerns the closed-loop latency of the ROS 2 / Gazebo stack. It is important to distinguish this *software* latency from the *robot-dynamics* lag of Sec. III-A: the empirically identified $\tau = 0.35\,\mathrm{s}$ in (3) is a physical PT_1 property of the UR5e end-effector chain (MoveIt 2 Servo solver, joint-PD tracking, simulated actuator inertia) that we deliberately included in the LQR plant model so that the controller anticipates it instead of fighting it. The Gazebo and ROS 2 message-passing pipeline introduces an *additional* cycle-to-cycle scheduling delay — physics step transport, the inverse-Jacobian solve, the EKF update, TD3 inference, and the ROS 2 callback queue — which is *not* part of the LQR plant model.

At a 30 Hz control loop the per-cycle budget is 33.3 ms; the LQR is robust to scheduling delays a few milliseconds in size by virtue of its sample period T_s alone, since T_s already enters the discretisation (5) of the dynamics. The residual policy, by contrast, is trained on simulator-time states and emits high-bandwidth corrections; when those corrections arrive late relative to the closed-loop physics, they appear in the recorded trajectory as the high-frequency over-correction transients ("jitter") visible in Scenario 2. We hypothesise—but have not isolated experimentally—that this latency-induced phase mismatch is a contributor to the residual's underperformance under jittered disturbance. A direct, per-cycle latency budget (physics, Servo, EKF, inference, callback queue) is left as a measurement exercise under the latency-aware action smoothing item in Sec. VIII-B.

TABLE III
SENSITIVITY OF HEAD-TO-HEAD PERFORMANCE TO ODE FRICTION COEFFICIENT μ ON SCENARIO 1.

μ	LQR win (%)	Hybrid win (%)	Δ_{win} (pp)	RMSE H/L
0.3	70	30	-40	0.1787/0.1981
0.5	70	30	-40	0.1741/0.2120
0.6	40	60	$+20$	0.1920/0.2162
0.8	50	50	0	0.2131/0.2032
1.0	30	70	$+40$	0.1826/0.1811

D. Sensitivity to contact friction

This experiment tests the robustness of the residual controller in the face of environmental instability. This is tested with the contact friction coefficient. All main results use $\mu = 0.6$, the Gazebo ODE default and a value consistent with steel-on-acrylic contact. To address robustness explicitly, we conducted (resp. are conducting) a sweep of $\mu \in \{0.3, 0.5, 0.6, 0.8, 1.0\}$ on Scenario 1, holding all other parameters fixed and running 10 paired trials per coefficient for both pure LQR and the 200 k Hybrid policy. Table III collects the metric of primary interest—the reward-superiority win-rate gap $\Delta_{\text{win}} = \text{win}_{\text{Hybrid}} - \text{win}_{\text{LQR}}$, where each win counts the paired seeds in which that controller's cumulative episode reward exceeds the other's—and the RMSE of each controller (Hybrid first, LQR second).

The data tells a more interesting story than the symmetric attenuation we had initially expected. Rather than peaking at the training-point $\mu = 0.6$ and falling off on either side, the win-rate gap exhibits an approximately monotone trend across the swept range: from $\Delta_{\text{win}} = -40$ pp at $\mu = 0.3$, through the original $+20$ pp at $\mu = 0.6$, to $+40$ pp at $\mu = 1.0$. Three observations follow.

Low friction breaks the residual's value-add: At $\mu \in \{0.3, 0.5\}$ the marble enters a slip-prone contact regime in which neither controller can mount a saturating rescue without losing traction; survival becomes the binding constraint on cumulative reward and the LQR's saturation envelope is the more reliable recovery. The residual, trained at $\mu = 0.6$, emits higher-bandwidth corrections that are out of distribution for the slipping plant and accelerate the loss-of-contact failure mode, costing the Hybrid ~ 40 pp of win-rate. Notably, on the seeds where the marble *does* survive the residual still produces a $\sim$ 10–18% lower RMSE than the LQR (0.1787 vs. 0.1981 at $\mu = 0.3$; 0.1741 vs. 0.2120 at $\mu = 0.5$), reinforcing the variance–survival decoupling we already observed in Scenario 2 (Sec. VI).

High friction unexpectedly favours the residual: At $\mu = 1.0$ the Hybrid wins the head-to-head by $+40$ pp despite an essentially tied RMSE (0.1826 vs. 0.1811 m). At sticky contact the rolling dynamics are well-conditioned and the marble's response to small plate-tilt corrections is nearly deterministic, so the residual's small-signal damping bias remains in-distribution and combines constructively with the LQR's saturation envelope across a wider set of seeds. We did not predict this asymmetry; it suggests that the operational envelope of the residual on this plant is broader on the high-friction side than on the slip side.

The training-point advantage is real but not the maximum: The $+20$ pp gap at $\mu = 0.6$ reproduces our Scenario 1 head-to-head result and confirms that the friction conditions of the original evaluation sit comfortably inside the residual's competence region; but it is neither the peak nor a sharp ridge. Across the four points $\mu \geq 0.6$ the residual is at parity or better, and across the two points $\mu < 0.6$ it underperforms only on the survival metric. The practical implication is that the residual is, in this respect, more robust to friction-coefficient drift than a hyperparameter-overfitting heuristic would predict—which is encouraging for sim-to-real transfer where the contact regime on a physical plate is essentially never the ODE default.

VIII. Conclusion and Future Work

We have presented a hybrid optimal-and-residual controller for a 6-DOF ball-and-plate stabilization task, characterized empirically the *operational envelope* of the residual contribution, and identified a structural failure mode of authority-scaling curricula—Curriculum Shock—that we believe applies to any off-policy residual policy gated by an authority threshold.

A. What we conclude

First, residual TD3 layered on a discrete-time LQR provably improves small-signal performance: in the nominal Scenario 1 the Hybrid controller wins the paired reward-superiority comparison on 60% of seeds (vs. LQR's 40%) and reduces RMSE by 1.1% at one third of the LQR's tilt authority. Second, the same residual cedes performance to the optimal baseline at extreme initial conditions because its training distribution under-samples that regime; this is a feature of any learned compensator operating outside its training support and is a clean characterization of the envelope, not a bug. Third, abrupt expansion of residual authority via curriculum induces irreversible critic divergence and policy collapse; we have measured the divergence at three orders of magnitude in critic loss and quantified the post-shock policy as effectively non-functional.

B. Future work

Shaped authority transitions: Replace the step transition by $\lambda(t) = \lambda^{(s)} + (\lambda^{(s+1)} - \lambda^{(s)})\, \sigma\big((t - t_0)/\tau_c\big)$, a logistic ramp with timescale τ_c chosen so that the increment per update step is small relative to the critic's estimation error.

Replay-buffer hygiene: At the transition, flush off-distribution transitions and warm-start the critic on the new authority bound before resuming actor updates.

Latency-aware action smoothing: Add an explicit second-order filter on $\boldsymbol{u}^{\text{RL}}$ to bound the closed-loop bandwidth below the ROS 2 / Gazebo round-trip latency.

Friction sweep (in progress)*:* Complete the sensitivity-to-friction study of Sec. VII-D and incorporate the results into the operational-envelope characterization.

Sim-to-real transfer: Deploy the Hybrid controller on a physical UR5e equipped with camera-based marble tracking and revisit the operational envelope under measurement noise and unmodeled actuator dynamics.

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